UNIT 3 — 20 QUESTIONS

Camera Calibration

Q41. Intrinsic camera parameters typically include:
- A. Rotation and translation only
- B. Focal length, principal point, and skew
- C. World coordinates of objects
- D. Frame rate of the video

Q42. Extrinsic camera parameters describe:
- A. The internal optics of the lens
- B. The camera's rotation and translation relative to the world
- C. The sensor's pixel size
- D. The distortion coefficients only

Q43. In calibration, focal length is often expressed in units of:
- A. Metres
- B. Pixels
- C. Radians
- D. Lux

Q44. Radial lens distortion causes straight real-world lines in an image to appear:
- A. Sharper
- B. Curved
- C. Duplicated
- D. Colour-shifted

Q45. "Barrel" distortion is characterised by straight lines that appear to:
- A. Bow inward toward the centre
- B. Bow outward away from the centre
- C. Remain perfectly straight
- D. Disappear at the edges

Q46. "Pincushion" distortion is characterised by straight lines that appear to:
- A. Bow outward away from the centre
- B. Bow inward toward the centre
- C. Rotate 90 degrees
- D. Become dashed

Q47. A commonly used calibration target for estimating camera parameters is a:
- A. Random noise pattern
- B. Checkerboard pattern
- C. Solid grey card
- D. Single point light source

Q48. Reprojection error in camera calibration measures the difference between:
- A. Two different camera models
- B. Observed image points and their reprojected estimated positions
- C. Focal length and principal point
- D. Two consecutive video frames

Q49. The orthographic projection camera model essentially discards:
- A. Colour information

B. Perspective/depth effects
C. Lens distortion only
D. Pixel resolution

Q50. The weak-perspective camera model is a reasonable approximation when:
A. The object is very close to the camera
B. Depth variation across the object is small relative to viewing distance
C. The lens has heavy distortion
D. The camera is moving rapidly

Q51. The general camera projection matrix P has dimensions:
A. 2×3
B. 3×3
C. 3×4
D. 4×4

Q52. Zhang's widely used calibration method requires:
A. A single image of a 3D calibration object
B. Multiple images of a planar pattern at different orientations
C. A stereo camera pair only
D. No calibration target at all

Q53. Intrinsic and extrinsic parameters can be recovered from a projection matrix using:
A. RQ (or QR) decomposition
B. Fourier transform
C. Watershed flooding
D. Mean-shift clustering

Q54. The principal point of a camera refers to:
A. The corner pixel of the sensor
B. The intersection of the optical axis with the image plane
C. The brightest pixel in the image
D. The centre of the calibration checkerboard

Q55. The skew parameter in the intrinsic matrix accounts for:
A. Radial lens distortion
B. Non-perpendicularity between the pixel axes
C. Frame rate mismatch
D. Colour channel misalignment

Q56. Tangential distortion in a lens typically arises due to:
A. Sensor overheating
B. Lens and sensor plane not being perfectly parallel (decentering)
C. Low ISO settings
D. Excessive image compression

Q57. Using more calibration images from varied viewpoints generally:
A. Increases reprojection error
B. Reduces reprojection error and improves accuracy
C. Has no effect on calibration quality
D. Only affects extrinsic parameters

Q58. The Direct Linear Transform (DLT) method is used to:
A. Estimate the projection matrix from 3D–2D point correspondences

B. Segment an image into regions
C. Compute optical flow vectors
D. Apply histogram equalization

Q59. The camera intrinsic matrix K primarily encodes:
A. World-to-camera rotation
B. Camera-internal parameters like focal length and principal point
C. The 3D position of tracked objects
D. Motion vectors between frames

Q60. Compared to the orthographic model, the full perspective camera model additionally accounts for:
A. Colour distortion
B. Depth-dependent foreshortening (perspective effects)
C. Frame compression
D. Sensor noise